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ORIGINAL ARTICLE

Autoregressive integrated moving average time series model for forecasting air pollution in Nanded city, Maharashtra, India

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Abstract

The present study assesses the time series data of air pollution in Nanded city since 2011 and is collected from source of Central Pollution Control Board, Government of India. In this study, autoregressive integrated moving average (ARIMA) time series forecasting approach is used for prediction of air pollution in Nanded city with two main locations. The main aim of this study is to forecast the air pollution for next 5 years by using ARIMA model with the air pollutants NO_x, SO_2, RSPM and SPM. The time series analysis is the predictive analytic technique gives us better results and this analysis is performed through free and open source software R. The prediction results obtained through ARIMA model highlight the increasing level of air pollution in Nanded city. In future, this study may be helpful to maintain the air pollutant within the permissible limit and in the development of smart city.

Keywords Air pollution · ARIMA · Time series · Forecasting · Nanded

Introduction

Now a days due to polluted atmosphere it is very difficult to get fresh and clean air from natural resources and it is observed that there is change in global warming and range of temperature. Air pollution is an exigent problem worldwide and the entire world is facing the hazard caused by it Throughout the world ambiance air pollution is recognized as major problem in living environment viz. harmful gases, dust, smokes in to environment. It became difficult to survive for humans, animals and plant (Zhang and Zhang 2017). Developing country like India, Air pollution is a big anxiety for India. Further, urban environment is polluted by several constituents like vehicles, improperly managed urbanization and increased unmanaged industrialization, it lead to make destructive consequence on urban ecosystem. Furthermore, health problems caused by air pollution affects the economic productivity of agricultural. Due to such polluted air there is possibility of environmental disaster which can damage the physical property of urban environment. Urban air quality is a matter of concern because of large number of vehicular emission which produces and increases the level of air pollutants viz. SPM, RSPM, SO_2, NO_x, other organic and inorganic including trace metal pollutants. The increasing number of vehicles and their use changes the composition ratio of pollutants. The atmospheric nature carries toxic species which may have detrimental effects on human health and ecosystems. Scientific evidence also shown that there is a strong relationship between SO_2 and NO_x pollutant with serious health issues (Wang et al. 2017). Some researchers studied the presence of hazardous chemical parameters in air and their effect on human health. It may cause high risk of asthma and other respiratory diseases. Also, air pollution may effect on lungs functionality as they suffered from respiratory inflammation and it may have a higher risk of cancer. It may damage the immune system, endocrine, reproductive systems and heart diseases (Dastoorpoor et al. 2018). Industrialization processes viz. fertilizer manufacturing, aluminum smelting and steel manufacturing generate the SO_2

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gas in environment in large amount geothermal activity like fossil fuels are the leading sources of sulphur dioxide (SO_2). Sulphate is another secondary air pollutant which produced by wood combustion, diesel and gasoline powered exhaust and road dust are responsible for the higher level of air pollutants in urban area. Furthermore, health problems caused by air pollution affects the economic productivity of agricultural and there is possibility of environmental disaster may damage the physical property of urban environment (Vanov et al. 2012).

Nitrogen Oxides (NO_x) is group of gases which is generated due to use of vehicle automobiles, power plants, processes used in chemical plant which leads to causes serious health issues. NO_x polluted air leads the problems like unhealthy to breathe, especially for children, the elderly persons, asthmatics and people with chronic obstructive pulmonary disease. The high amount of NO_x in atmosphere causes leaf damage and reduces the growth of plant. Further, it makes vegetation more vulnerable to disease and frost damage (Wang et al. 2017; Vanov et al. 2012).Suspended Particulate Matter (SPM) is one of the major pollution particle which is responsible for deaths in urban environment. A polluted air with high amount of SPM particles can cause respiratory infection in children's health and for adult people SPM causes diseases like cardiovascular diseases, lung cancer, and chronic respiratory diseases. The SPM particles mostly consist of dust, fumes, mist and smoke particles in air. Due to high amount of diesel consumption by vehicles chemical components viz. lead, nickel, arsenic etc. were formed and those are responsible for breath problem, lungs issues and respiratory problems. Epidemiological evidence has shown the impact of SPM on global environment (Su et al. 2016). The major source of respirable suspended particulate matter (RSPM) is combustion sources of domestic heating, industrial emissions, vehicle emissions, wind-blown dust and outdoor burning. The RSPM particles small in nature and they are effectively act as a gas. The RSPM particles can penetrate deep into the lungs and results as serious health issues such as asthma attacks, high blood pressure, heart attack, strokes and premature death (Dastoorpoor et al. 2018).

Time series model comprises methods for analyzing time series data in order to extract the meaningful statistics and other characteristics of data. Generally, time series models were used for the forecasting the values based on the previous observed values. The time series analysis is evaluated to understand the structure of air pollution parameters and to predict future possible values, trend and seasonality pattern in nature. Georgieva (2014) used time series analysis and univariate stochastic seasonal autoregressive integrated moving average (SARIMA) models. Johnson power transformation for variance stabilizing of the data and model selection by using Bayesian information criterion is used. The SARIMA models result produced good fitting performance with short term prediction for 72 h ahead. The statistical approach allowed building noncomplex model which was effective for short term air pollutant forecasting to warn urban environmental behavior. Kadiyala and Kumar (2014) demonstrated the methodology for the development and validation of multivariate time series transfer function models (ARMAX/ARIMAX) for the in bus contaminant concentrations of carbon dioxide and carbon monoxide using available software. Yi and Chan (2015) investigated the effects of heat and cold temperature on daily mortality in Hong Kong for the time period of 2002–2011. Findings of their study shown that, it has implications for the development of intervention strategies and it may be useful for protection in the extremely cold and hot temperatures environment. Bhardwaj and Pruthi (2016) study deals with the estimation of regression coefficient, Hurst exponent, Fractal Dimension and Predictability Index of air pollution particles such as NO, NO_2, NO_x, SO_2, $PM_{2.5}$ and the different weather conditions in the Dwarka area adjoining Indira Gandhi International Airport, Delhi. The Brownian time series method is used for the prediction air pollution. It is observed that the odd–even scheme have less impact on some air pollutant parameters. Due to air pollution parameters the trend cannot forecasted for identification of behavior and pattern in the study area. Yu (2016) dealt with the time series air quality monitoring data and analyzed data through exploratory analysis. Guo et al. (2017) had studied the estimation of associations between ambient air pollutant and mortality outcomes in Shenzhen city for the period of 2012–2014. The results showed that nitrogen dioxide (NO_2)-induced excess risks (ER) of total non-accidental mortality and cardiovascular mortality were significantly increased. The purpose behind using time series analysis was to generalize additive Poisson regression models to estimate the associations between air pollutants and mortalities. Further, they also performed sensitivity analysis to explore the robustness of the models. Dimakopoulou et al. (2017) proposed the spatio-temporal land use regression (LUR) models to account for spatial variability. The time series analysis is used to access the effects of short term exposures to PM_{10} and NO_2 the results demonstrated using spatio-temporal models in time series analysis for short-term air pollution effects to account for spatial variability in addition to the

temporal. Kadilar and Kadilar (2017) analyzed the level of SO_2 in Aksaray city based on the database monitored by air quality monitoring station of Turkey. The SARIMA approach is used for forecasting the levels of SO_2 and result provided the reliable and satisfactory predictions for air quality parameter. The main aim of this study is to identify the variation of SO_2, NO_x, RSPM and SPM parameters present in the air and to identify the trend of air pollution parameters in respective locations of Nanded city. The predicted results will be helpful in planning to maintain the air pollution level and other characteristics of data.

Study area

Nanded city having multifunctional environment which is district headquarter. In recent year Nanded is spread out widely with complex urban structure. Central part of Nanded has faced unusual social and physical transformation changes. These changes played a vital role in social, economic and cultural development of Nanded and its surrounding regions. One of the major cities in Marathwada Region in Maharashtra is Nanded which is located on the bank of Godavari River. Nanded is situated at latitude of 19.15 and longitude of 77.30. The total population of Nanded Waghala City Municipal Corporation was 550,564 (Census report of India 2011; Kulkarni et al. 2017) (Fig. 1).

Materials and methods

To forecast the level of air pollution parameters assessment in Nanded city the data has been collected (2011–2016) from Maharashtra pollution control board (MPCB 2017) with the frequency of 2 days in a week. Two air pollution counting mechanism are nearby in study area located respectively at Vazirabad and at Ganesh Nagar. The aim of the present study is to explore descriptive analysis and to forecast air pollution data Autoregressive Integrated Moving Average (ARIMA) time series approach for next 5 years. The ARIMA method is an iterative, exploratory, process which consist of three major steps as identification, estimation and diagnostic. The ARIMA consist of auto-regressive component (AR) denoted as p, integrated component (I) as d, and moving average (MA) as q. Auto regressive components represents the lingering effects of previous observations. Integrated component (I) represents trends, including seasonality. Further, Moving Average represents lingering effects of previous random shocks or error. While applying ARIMA model it is important to maintain the order of each component. In general the order of component starts with the small integer values (as 0, 1 or 2). The component order is basically set to find out best parsimonious model with estimated parameters. The function work of Time series analysis is based on the mathematical function which explain the nature of data in such way that prediction, monitoring, or control can easily occur in systematic pattern. The general mathematical form of ARIMA model is represented in Eqs. (1–5). Specifically, ARIMA (p, 1, q) has been obtained from ARIMA (p, d, q) model for d = 1 with difference of Y_t further, it is represented in equation following equations (Naveen and Anu 2017):

$$\Delta_d Y_t = W_t \quad \text{or} \quad W_t = Y_t - Y_{t-1}. \tag{1}$$

$$\begin{aligned} \text{When} \quad W_t &= Y_t - Y_{t-1} \\ W_{t-1} &= Y_{t-1} - Y_{t-2} \\ W_{t-2} &= Y_{t-2} - Y_{t-3}. \end{aligned}$$

Similarly, when: $t = t - p$ then $W_{t-p} = Y_{t-p} - Y_{t-p-1}$.

$$\begin{aligned} W_t = \Phi_1 W_{t-1} + \Phi_2 W_{t-2} + \cdots + \Phi_p W_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} \\ - \theta_2 \varepsilon_{t-2} - \cdots \theta_q \varepsilon_{t-q}. \end{aligned} \tag{2}$$

After subtraction we get the following Eqs. (3, 4)

$$\begin{aligned} W_t = \Phi_1 \left(Y_{t-1} - Y_{t-2}\right) + \Phi_2 \left(Y_{t-2} - Y_{t-3}\right) + \cdots \\ + \Phi_p \left(Y_{y-p} - Y_{t-p-1}\right) + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \cdots - \theta_q \varepsilon_{t-q} \end{aligned} \tag{3}$$

$$\begin{aligned} Y_t - Y_{t-1} = \Phi_1 Y_{t-1} - \Phi_1 Y_{t-2} + \ldots + \Phi_p Y_{t-p} - \Phi_p Y_{t-p-1} \\ + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \cdots - \theta_q \varepsilon_{t-q}. \end{aligned} \tag{4}$$

The final equation becomes (Eq. 5):

$$\begin{aligned} Y_t = \left(1 + \Phi_t\right) Y_{t-1} + \left(\Phi_2 - \Phi_1\right) Y_{t-2} + \left(\Phi_3 - \Phi_2\right) Y_{t-3} + \cdots \\ + \left(\Phi_p - \Phi_{p-1}\right) Y_{t-p} + \Phi_p - Y_{t-p-1} + \varepsilon_t - \theta_1 \varepsilon_{t-1} \\ - \theta_2 \varepsilon_{t-2} - \cdots - \theta_q \varepsilon_{t-q}. \end{aligned} \tag{5}$$

In this study, the analysis is performed through MS Excel and R software 3.4.1.

Table 1 Descriptive statistics of parameters of the study area

Parameters	Vazirabad					Ganesh Nagar				
	Average	Min	Max	Sd	Sem	Average	Min	Max	Sd	Sem
SO_2	32.04	4	66.12	11.28	1.32	22.78	2.6	41.62	8.42	0.99
NO_x	28.04	4	64.37	9.33	1.10	23.94	2	77.44	10.23	1.20
RSPM	93.73	3.55	237.25	63.27	7.45	62.56	2.77	122.1	43.17	5.08
SPM	206.78	8.44	329.62	98.676	11.62	123.94	6.6	181	47.78	5.63

Table 2 Predicted average range of air pollution for next 5 years (Vazirabad)

Year	SO_2	RSPM	NO_x	SPM
2017	15.91–56.66	1.75–244.83	13.91–47.91	59.41–420.833
2018	13.66–59.66	0–288.83	12.33–49.58	36.33–493.41
2019	10.5–62.75	0–319.08	9.5–52.25	0–523.33
2020	7.5–65.83	0–348.33	7.33–54.41	0–560
2021	4.83–68.25	0–372.25	5.33–56.33	0–590

Table 3 Predicted average range of air pollution for next 5 years (Ganesh Nagar)

Year	SO_2	RSPM	NO_x	SPM
2017	11.5–41.16	0–127	6.41–47.83	66.91–232.33
2018	9.08–44.16	0–170.5	5.66–48.58	48.75–254.75
2019	5.33–47.7	0–197.75	1.83–52.58	48.75–277.33
2020	3–50	0–222.33	0–54.08	7.58–249.25
2021	1–52.41	0–246.16	0–56.58	0–310.41

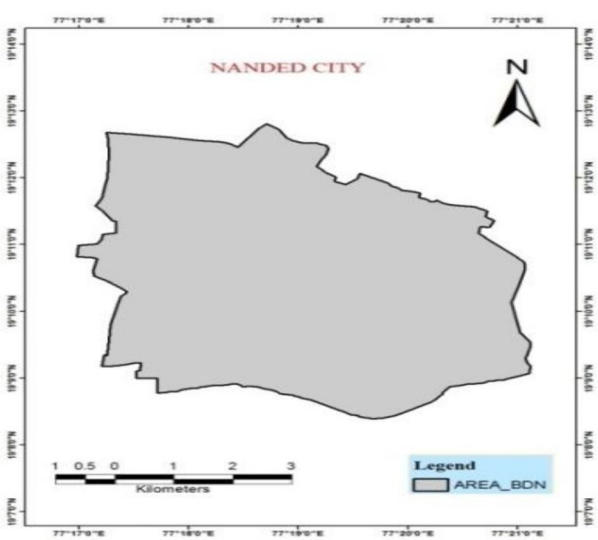

Fig. 1 Nanded city

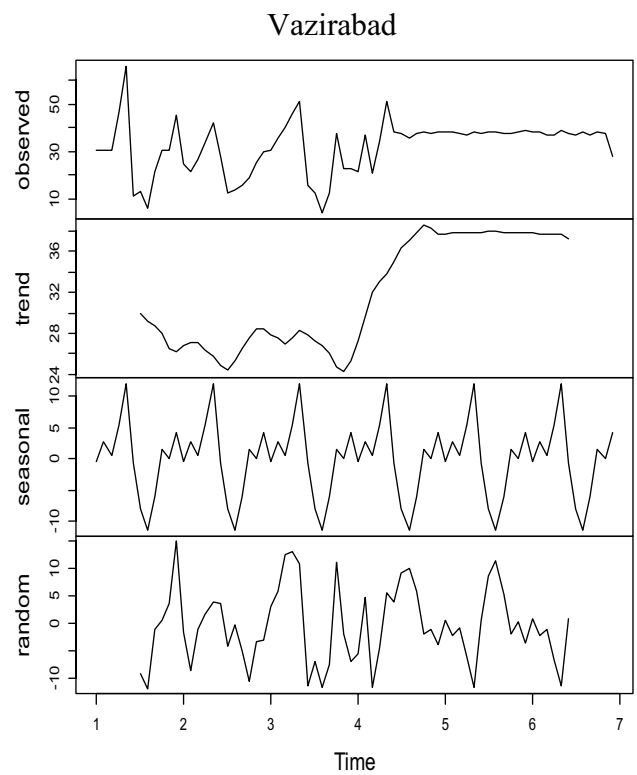

Fig. 2 Decomposition of additive time series of SO_2

Results and discussion

The present study analyzed the air pollution of two different places of Nanded city based on Central pollution control-board regulation (CPCB 2013). The investigation results that variation in air pollution is present during the summer and winter. The summary of air pollution is presented in Table 1.

It shows the descriptive levels of pollutants in the studied region. The average modeled concentrations of SO_2 and NO_x for Vazirabad 32.04 and 28.04 μg/m^3. The average concentration of RSPM and SPM for Vazirabad is 93.73

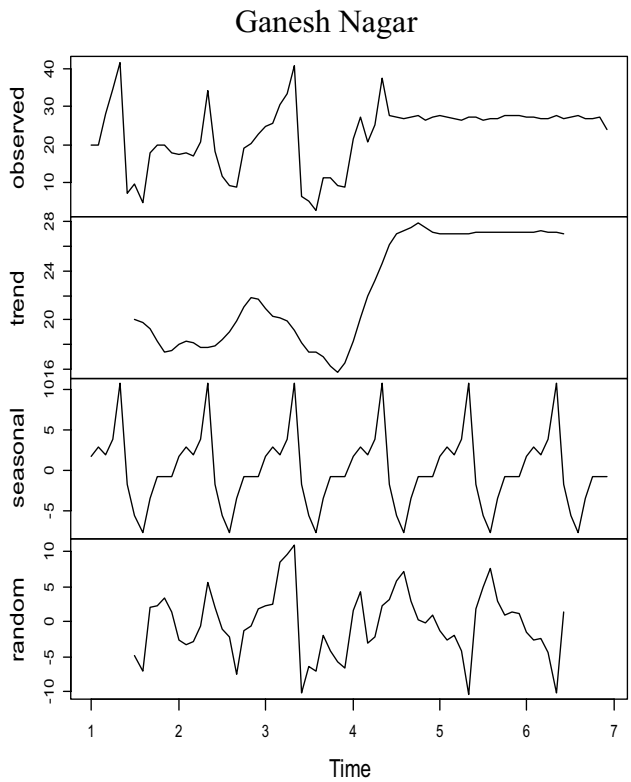

Fig. 3 Decomposition of additive time series of SO_2

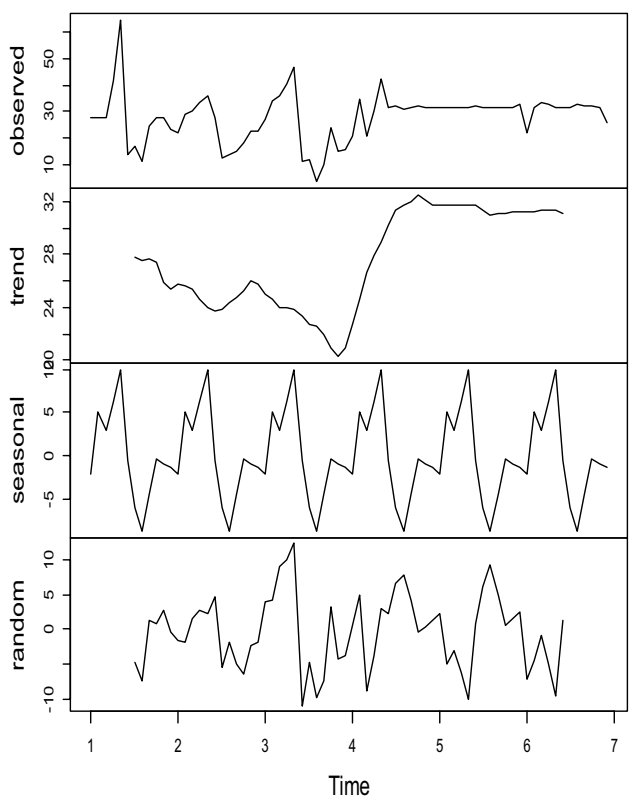

Fig. 4 Decomposition of additive time series of NO_x

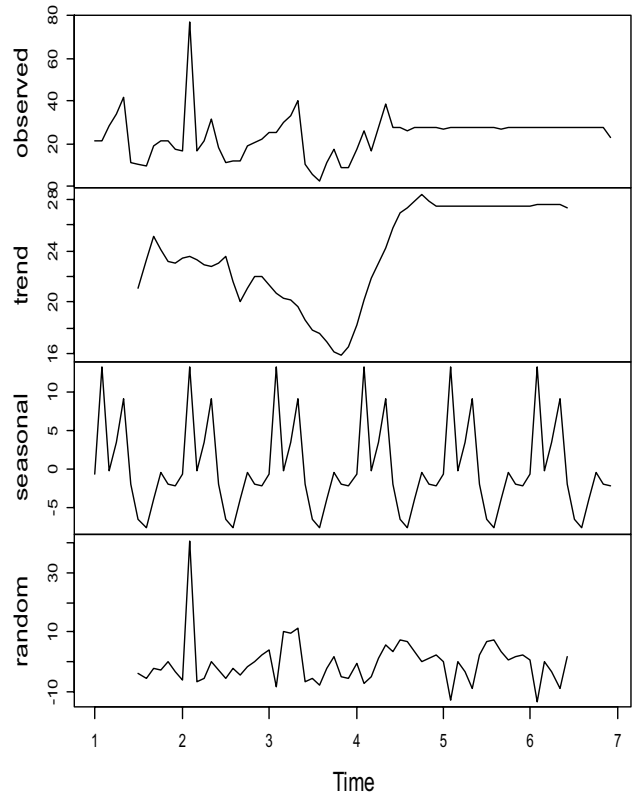

Fig. 5 Decomposition of additive time series of NO_x

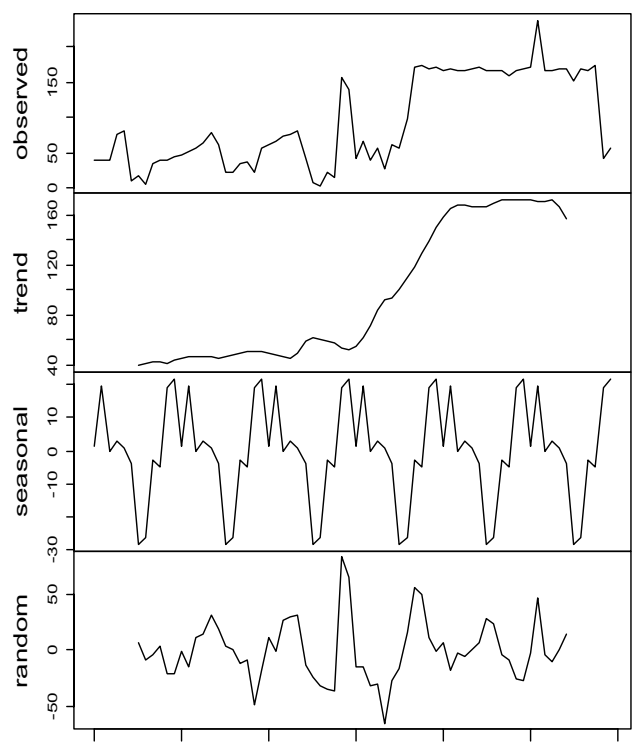

Fig. 6 Decomposition of additive time series of RSPM

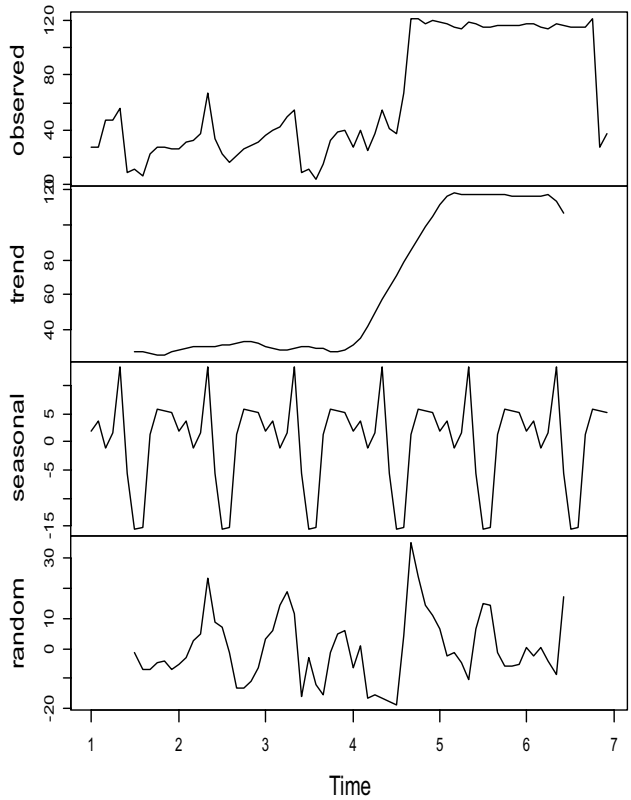

Fig. 7 Decomposition of additive time series of RSPM

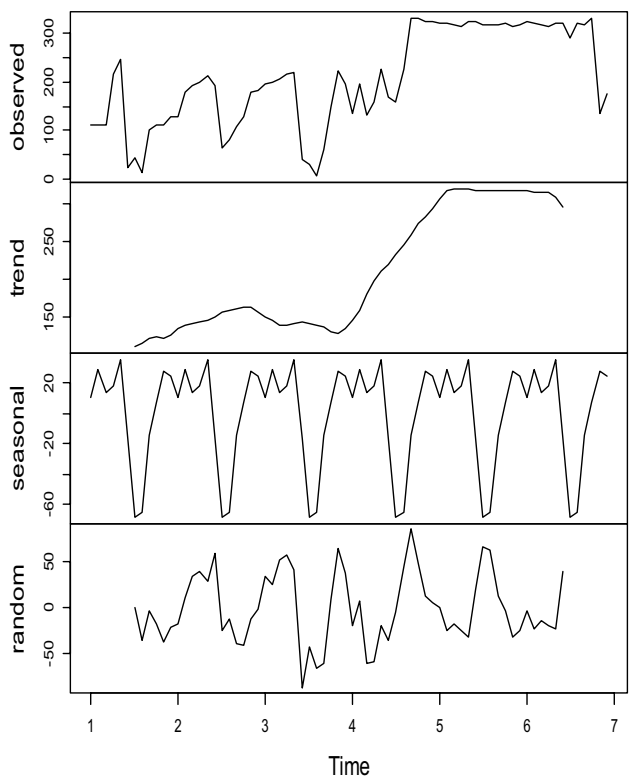

Fig. 8 Decomposition of additive time series of SPM

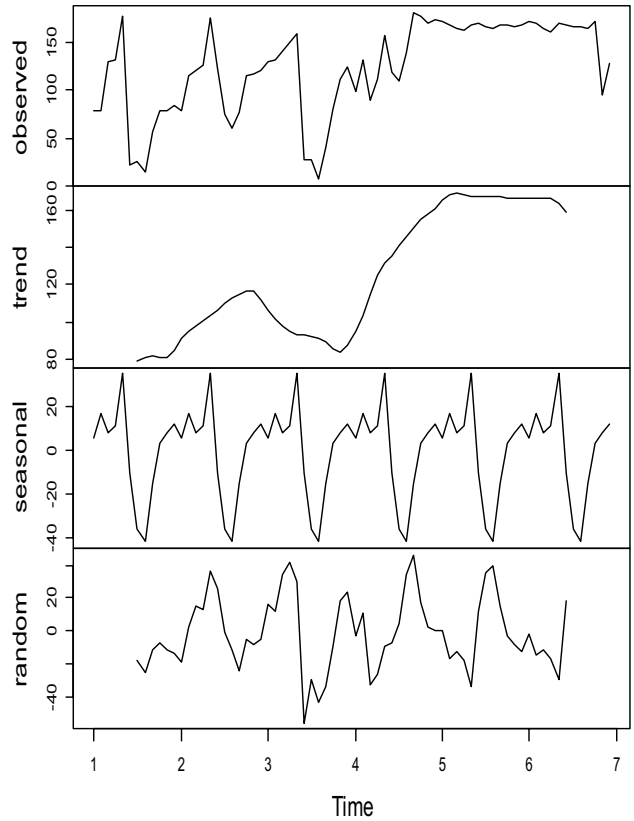

Fig. 9 Decomposition of additive time series of SPM

and 206.78 μg/m^3 respectively. The average modeled concentrations of SO_2 and NO_x for Ganesh Nagar 22.78 and 23.94 μg/m^3 respectively. The average modeled concentrations of RSPM and SPM for Ganesh Nagar 62.56, 123.94 μg/m^3 respectively. The maximum modeled concentrations of SO_2, NO_x, RSPM, SPM for Vazirabad 66.12, 64.37, 237.25, 329.62 μg/m^3 with standard deviation value 11.28, 9.33, 63.27, 98.67 respectively. The maximum modeled concentrations of SO_2, NO_x, RSPM and SPM for Ganesh Nagar ranged 41.62, 77.44, 122.1, 181 μg/m^3 with standard deviation 8.42, 10.23, 43.17, 47.78 respectively. The permissible limit for SO_2 and NO_x is 80 μg/m^3. For RSMP the standard limit is 100 μg/m^3. The air pollution was determined by taking the mean value of the selected observations over 72 months. This enables us to compare air pollution between two points of Nanded city. Based on the collected information it is observed that SO_2, $NO_{x,}$ RSPM and SPM are the major air pollutant in Nanded city. Also, Vazirabad regions

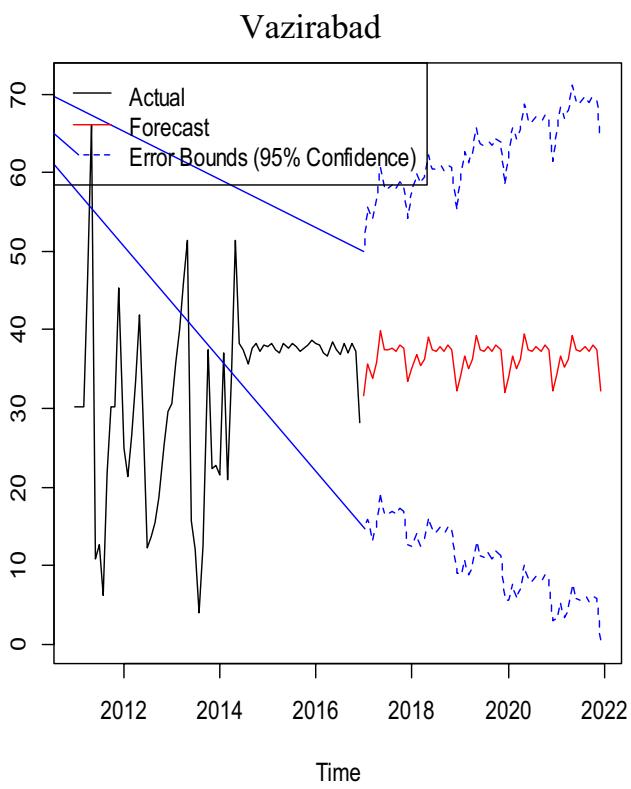

Fig. 10 Representation of forecasted values of SO_2

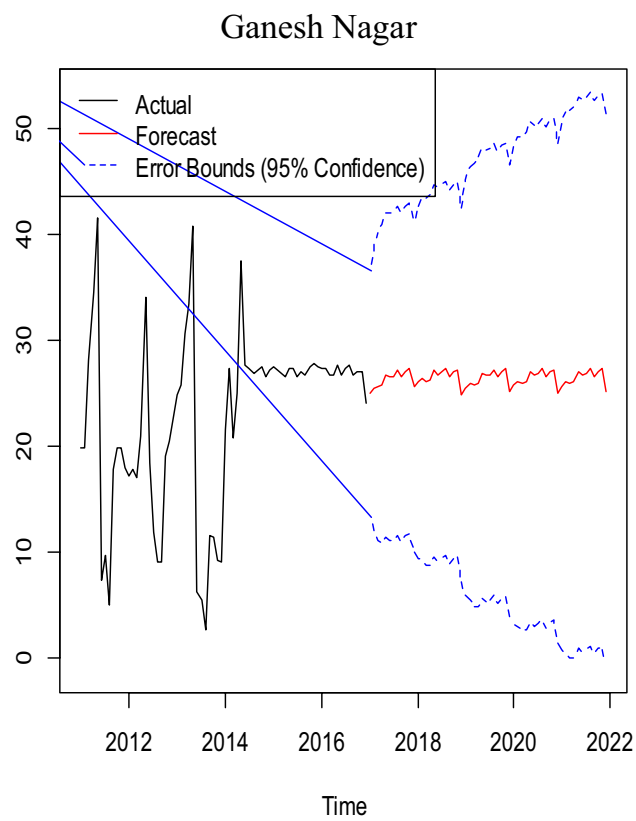

Fig. 11 Representation of forecasted values of SO_2

atmospheric condition is found to be more polluted as compared with Ganesh Nagar. It is observed due to heavy traffic flow and commercial movement in Vazirabad. Also, it is found that most of the part of Ganesh Nagar is residential area; it may lead to least pollutions compared with Vazirabad region. It is also found that, during November and April month air pollution is more as compared with rest of the month.

Tables 2 and 3 represents the prediction values of air pollution parameter for next 5 years of Vazirabad and Ganesh Nagar respectively. The forecasted result reveals that possible range of these parameters; RSPM and SPM air pollutants are overstepping the permissible limit in both region. Further, RSPM and SPM pollutants are overstepping the permissible limit in Vazirabad region more than Ganesh Nagar. Also, SO_2 and NO_x are lies in within the permissible limit.

Conclusion

In the present study, the descriptive statistics and time series is performed to identify the characterization air pollution. The investigation reveals that, during the summer and winter season significant variation in air pollution. Figures 2, 3, 4, 5, 6, 7, 8 and 9 represents the decomposition of additive time series of SO_2, NO_x, RSPM and SPM parameters respectively. Figures 10, 11, 12, 13, 14, 15, 16 and 17 represented forecasted values of air pollution parameters of SO_2, NO_x, RSPM, and SPM of Vazirabad and Ganesh Nagar. The forecasting values reveals that the air pollutant RSPM and SPM are overstepping the permissible limit which needs to be control. Every year the significant variation of pollutant concentration is identified

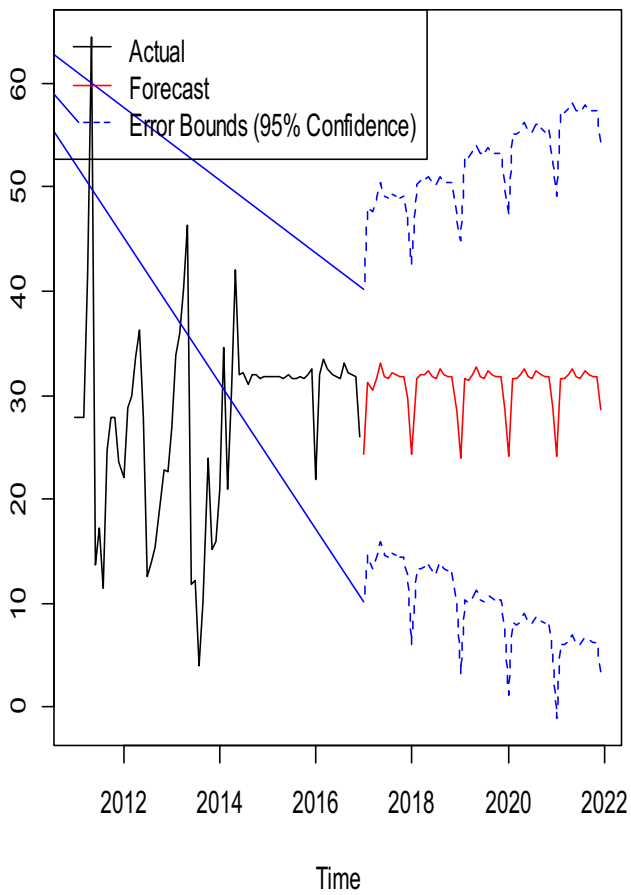

Fig. 12 Representation of forecasted values of NO_x

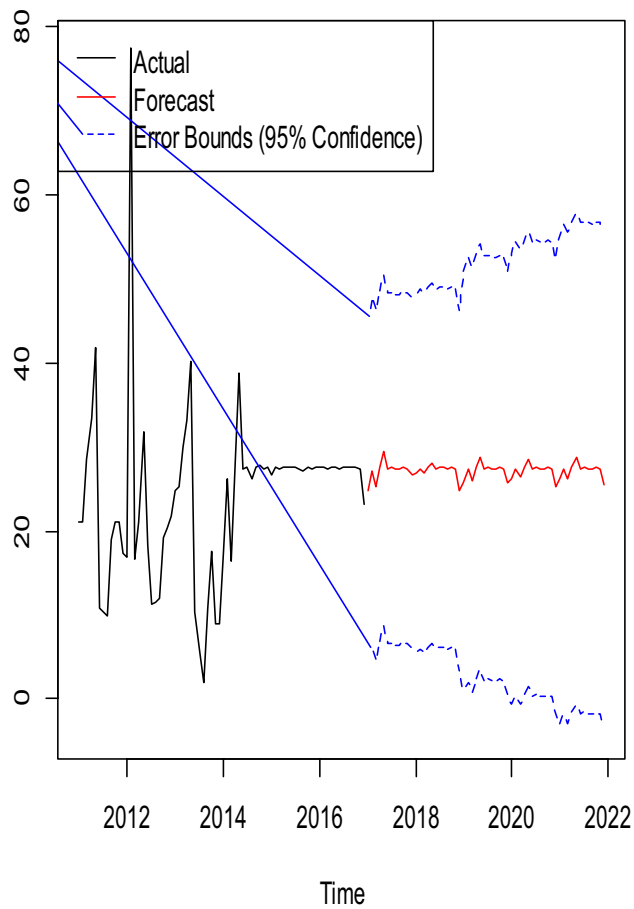

Fig. 13 Representation of Forecasted values of NO_x

during November–April. Also, Vazirabad regions atmospheric condition is found to be more polluted due to heavy traffic flow and commercial movement as compared with Ganesh Nagar. The major source of Nitrogen dioxide is fuel consumption which forms irritant gas. In atmosphere, nitrogen is present in the form of nitric oxide. Further, when it get mixed with oxygen it creates nitrogen dioxide (NO_2). The Nitrogen dioxide and nitric oxide are referred to together as oxides of nitrogen (NO_x). In SPM chemical component like lead, nickel, arsenic etc. are formed due to high amount of diesel consumption through vehicle use. These particles are responsible for breath problem, lungs issues and respiratory problems. Industrialization processes viz. fertilizer manufacturing, aluminum smelting and steel manufacturing generate the SO_2 gas in environment in large amount geothermal activity like fossil fuels are the leading sources of sulphur dioxide (SO_2). To control the effect of air pollution remedies need to be applied

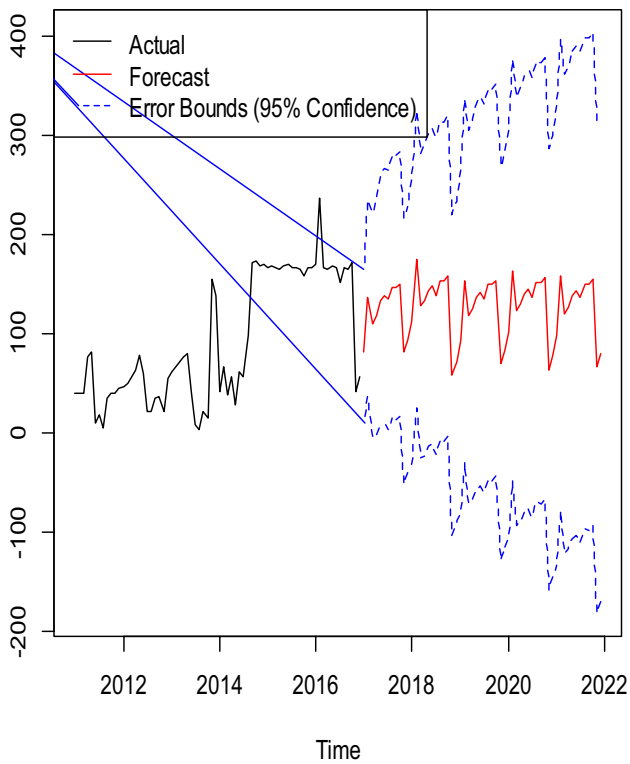

Fig. 14 Representation of forecasted values of RSPM

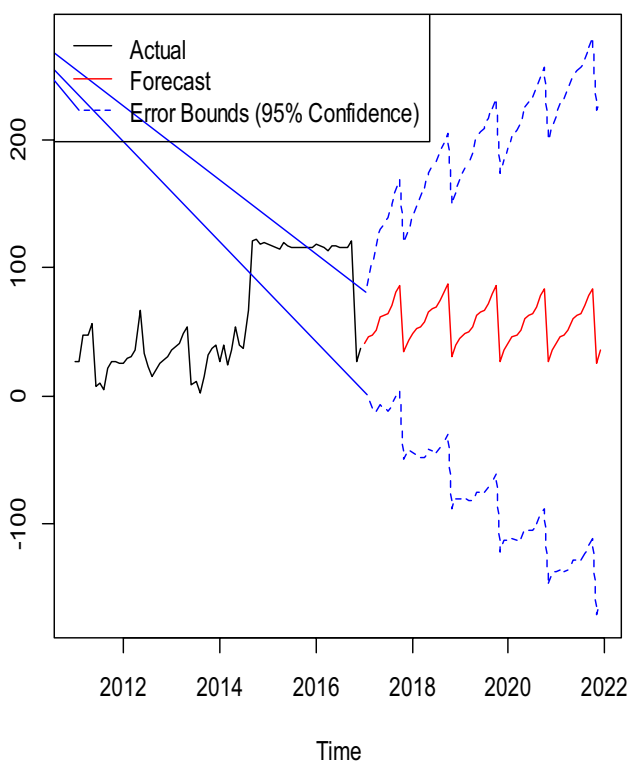

Fig. 15 Representation of forecasted values of RSPM

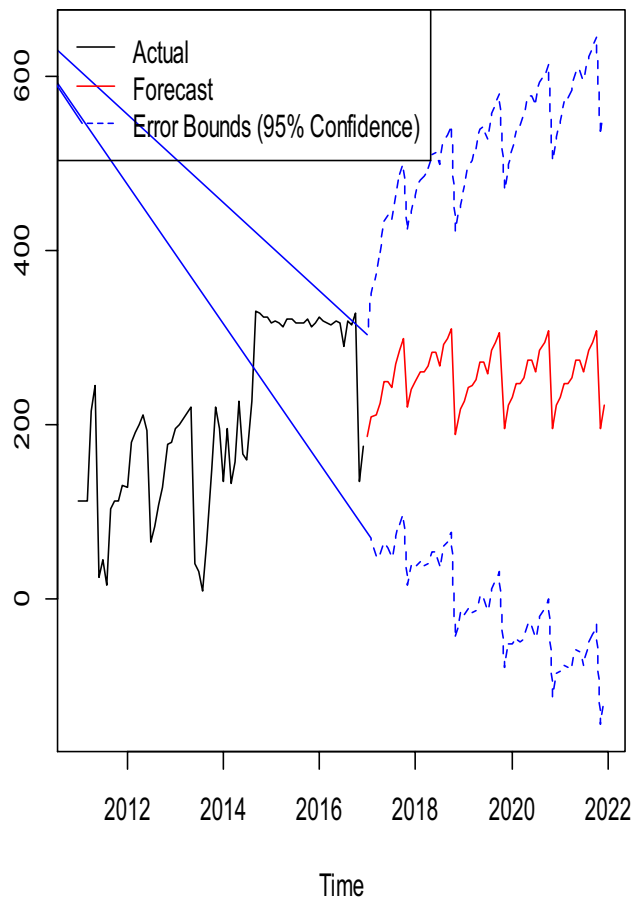

Fig. 16 Representation of forecasted values of SPM

such as protection of urban green areas i.e. urban forest which maintain the quality of air. Further, they can reduce the carbon-dioxide (CO_2) level and increases the oxygen (O_2) level in urban environment. It is necessary to create green belts in urban areas with strict restrictions. These is need of reengineering in automobile sector in such a way that their emission causes minimum pollution. Government should encourage people to use public transport and should avoid vehicles for short distance. Further, New policies need to be design with old policies up-gradation for green space management. Technical devices should be used more for controlling air pollution. To control the air pollution we highly recommend to plant more pollution tolerance trees which will produces pure healthy air in the regions. This study will be helpful to overcome the air pollution problem. This study helps to overcome air pollution problems of Nanded city. It helps to local planning and development authorities to reduce the air pollution Nanded city.

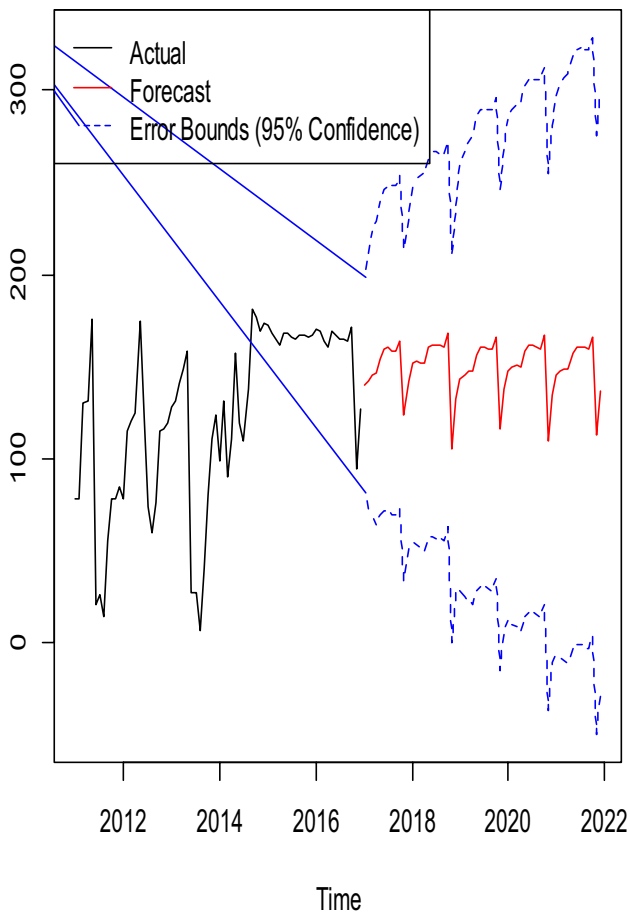

Fig. 17 Representation of Forecasted values of SPM

References

Bhardwaj R, Pruthi D (2016) Time series and predictability analysis of air pollutants in Delhi. In: 2nd international conference next generation computing technologies, IEEE, pp 553–560

Census report of India (2011) https://www.census2011.co.in/census/district/349-nanded.html

CPCB (2013) http://cpcb.nic.in/air-pollution/

Dastoorpoor M, Idani E, Goudarzi G, Khanjani N (2018) Acute effects of air pollution on spontaneous abortion, premature delivery, and stillbirth in Ahvaz, Iran: a time-series study. Environ Sci Pollut Res 25:5447–5458

Dimakopoulou K, Gryparis A, Katsouyanni K (2017) Using spatio-temporal land use regression models to address spatial variation in air pollution concentrations in time series studies. Air Qual Atmos Health 10(9):1139–1149

Gocheva-Ilieva SG, Ivanov AV, Voynikova DS, Boyadzhiev DT (2014) Time series analysis and forecasting for air pollution in small urban area: an SARIMA and factor analysis approach. Stoch Env Res Risk Assess 28(4):1045–1060

Gocheva-Ilieva SG, Ivanov AV, Voynikova DS, Boyadzhiev DT (2014) Time series analysis and forecasting for air pollution in small urban area: an SARIMA and factor analysis approach. Stoch Environ Res Risk Assess 28(4):1045–1060

Guo Y, Ma Y, Zhang Y, Huang S, Wu Y, Yu S, Cheng J (2017) Time series analysis of ambient air pollution effects on daily mortality. Environ Sci Pollut Res 24(25):20261–20272

Kadilar G, Kadilar C (2017) Assessing air quality in Aksaray with time series analysis. AIP Conf Proc 1833(1):020112

Kadiyala A, Kumar A (2014) Multivariate time series models for prediction of air quality inside a public transportation bus using available software. Environ Prog Sustain Energy 33(2):337–341

Kulkarni G, Nilesh D, Parag B, Wasnik P, Hambarde K, Tamsekar, Kamble V, Bahuguna V, Bahuguna V (2017) Effective use of GIS based spatial pattern technology for urban greenery space planning: a case study for Ganesh Nagar area of Nanded city. In: Proceedings of 2nd international conference on intelligent computing and applications. Springer, Singapore, pp 123–132

MPCB (2017) http://mpcb.gov.in/envtdata/demoPage1.php

Naveen V, Anu N (2017) Time Series Analysis to Forecast Air Quality Indices in Thiruvananthapuram District, Kerala, India. J Eng Res Appl 7(6):66–84

Su C, Breitner S, Schneider A, Liu L, Franck U, Peters A, Pan X (2016) Short-term effects of fine particulate air pollution on cardiovascular hospital emergency room visits: a time-series study in Beijing, China. Int Arch Occup Environ Health 89(4):641–657

Vanov A, Voynikova D, Gocheva-Ilieva S, Boyadzhiev D (2012) Parametric time-series analysis of daily air pollutants of city of Shumen, Bulgaria. AIP Conf Proc 1487(1):386–396

Wang KW, Deng C, Li JP, Zhang YY, Li XY, Wu MC (2017) Hybrid methodology for tuberculosis incidence time-series forecasting based on ARIMA and a NAR neural network. Epidemiol Infect 145(6):1118–1129

Yi W, Chan AP (2015) Effects of temperature on mortality in Hong Kong: a time series analysis. Int J Biometeorol 59(7):927–936

Yu C (2016) Research of time series air quality data based on exploratory data analysis and representation. In: Fifth international conference on agro-geoinformatics, IEEE, pp 1–5

Zhang H, Zhang S (2017) Forecasting of particulate matter time series using wavelet analysis and wavelet-ARMA/ARIMA model in Taiyuan, China. J Air Waste Manag Assoc 67:776–788